Research Paper

Instructor Performance and Course Quality Evaluation on EduPro

Abstract

Online education platforms rely heavily on instructor quality and course design to maintain learner satisfaction and platform credibility. This project develops a data-driven framework to evaluate instructor performance, course quality, and the relationship between teaching experience and learner satisfaction. Interactive dashboards and analytical techniques are used to identify top-performing instructors, high-quality courses, and improvement opportunities.

---

1. Introduction

Online learning has transformed education by providing learners with flexible access to quality content. However, maintaining consistent instructional quality remains a major challenge.

EduPro aims to improve educational outcomes by evaluating instructor effectiveness using data analytics.

This project analyzes instructor ratings, course ratings, teaching experience, expertise, and enrollments.

---

2. Problem Statement

EduPro currently lacks an objective method to evaluate instructors.

Major questions include:

- Which instructors perform the best?
- Does teaching experience improve ratings?
- Which expertise areas perform better?
- Does instructor quality affect enrollments?
- Which course categories need improvement?

---

3. Objectives

The objectives are:

- Analyze instructor performance.
- Evaluate course quality.
- Measure instructor experience impact.
- Identify top instructors.
- Compare expertise areas.
- Build an interactive dashboard.

---

4. Dataset Description

The project uses four datasets.

Teachers

- TeacherID
- TeacherName
- Age
- Gender
- Expertise
- YearsOfExperience
- TeacherRating

Courses

- CourseID
- CourseName
- CourseCategory
- CourseLevel
- CourseRating

Users

- UserID
- UserName
- Age
- Gender

Transactions

- TransactionID
- UserID
- TeacherID
- CourseID

---

5. Methodology

The following methodology was adopted.

Data Collection

Excel dataset provided by EduPro.

Data Cleaning

- Removed duplicates
- Checked missing values
- Converted data types
- Standardized categorical values

Data Integration

Merged all datasets using

- TeacherID
- CourseID
- UserID

Exploratory Data Analysis

Performed

- Descriptive statistics
- Correlation analysis
- Group analysis
- Visualization

Dashboard Development

Developed an interactive Streamlit dashboard with filters and KPIs.

---

6. Key Performance Indicators

- Average Teacher Rating
- Average Course Rating
- Total Teachers
- Total Courses
- Total Students
- Total Enrollments

---

7. Exploratory Data Analysis

The analysis includes:

Instructor Rating Distribution

Understanding the spread of instructor ratings.

Experience vs Rating

Analyzing whether experienced teachers receive better ratings.

Course Category Analysis

Comparing average ratings across different categories.

Expertise Analysis

Evaluating instructor performance by expertise.

Enrollment Analysis

Studying instructor popularity through enrollments.

Correlation Analysis

Measuring relationships between

- Experience
- Teacher Rating
- Course Rating

---

8. Dashboard Features

The dashboard includes:

- Sidebar Filters
- KPI Cards
- Scatter Plot
- Leaderboard
- Heatmap
- Pie Chart
- Course Analysis
- Download Dataset

---

9. Results

The analysis identified:

- High-performing instructors consistently receive better ratings.
- Course ratings vary by category.
- Teaching experience positively influences ratings up to a certain level.
- Some expertise domains outperform others.
- High-rated instructors generally attract more enrollments.

---

10. Recommendations

EduPro should:

- Provide training for lower-performing instructors.
- Encourage knowledge sharing among instructors.
- Improve low-rated course categories.
- Introduce regular instructor performance evaluations.
- Use dashboard analytics for decision-making.

---

11. Conclusion

This project successfully evaluates instructor performance and course quality using data analytics.

The Streamlit dashboard provides an interactive platform for monitoring instructor effectiveness and course performance.

The project enables EduPro to improve educational quality through evidence-based decision-making.

---

12. Future Scope

Future improvements include:

- Machine Learning prediction models
- Student feedback sentiment analysis
- Instructor recommendation system
- Real-time analytics dashboard
- Performance forecasting

---

13. References

1. Pandas Documentation
2. Streamlit Documentation
3. Plotly Documentation
4. Seaborn Documentation
5. Python Documentation